

R markdown for Supplementary PDF

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Data loading and set-up

Load in simulated data from the `simulate_data.R` file. This data was simulated for this manuscript. The true relationship is that age has a linear relationship and BMI has a U-shaped relationship.

We will assume that the data has been appropriately examined, plotted and cleaned.

```
# load in the simulated data
source("simulate_data.R")

# Set up data distribution information for rms. These two lines are standard
# when using the rms package
dd <- datadist(data)
options(datadist = "dd")
```

Data summary and demographics table

This is using `table1()` from the `table1` package to give a summary of the simulated data.

```
table1(~age + bmi + sex + smoking + as.factor(majorcomplication), data = data)
```

	Overall
	(N=5000)
age	
Mean (SD)	49.9 (12.1)
Median [Min, Max]	50.1 [7.40, 90.2]
bmi	
Mean (SD)	25.0 (3.96)
Median [Min, Max]	25.0 [11.2, 39.2]
sex	
Female	2467 (49.3%)
Male	2533 (50.7%)
smoking	
Never	1678 (33.6%)
Former	1677 (33.5%)
Current	1645 (32.9%)
as.factor(majorcomplication)	
0	4169 (83.4%)
1	831 (16.6%)

Model building and basic rmsMD outputs

```
# Fit logistic regression model using restricted cubic splines
fit <- lrm(majorcomplication ~ rcs(age, 3) + rcs(bmi, 3) + sex + smoking,
  data = data,
  x = TRUE, y = TRUE)

# note, x = TRUE, y = TRUE is recommended for lrm and cph models to allow subsequent
# Likelihood Ratio (LR) tests to be performed

# Check the model fit and diagnostics including number of observations and events
# Note the spline term coefficients (age, age', bmi, bmi') are difficult to interpret
fit
```

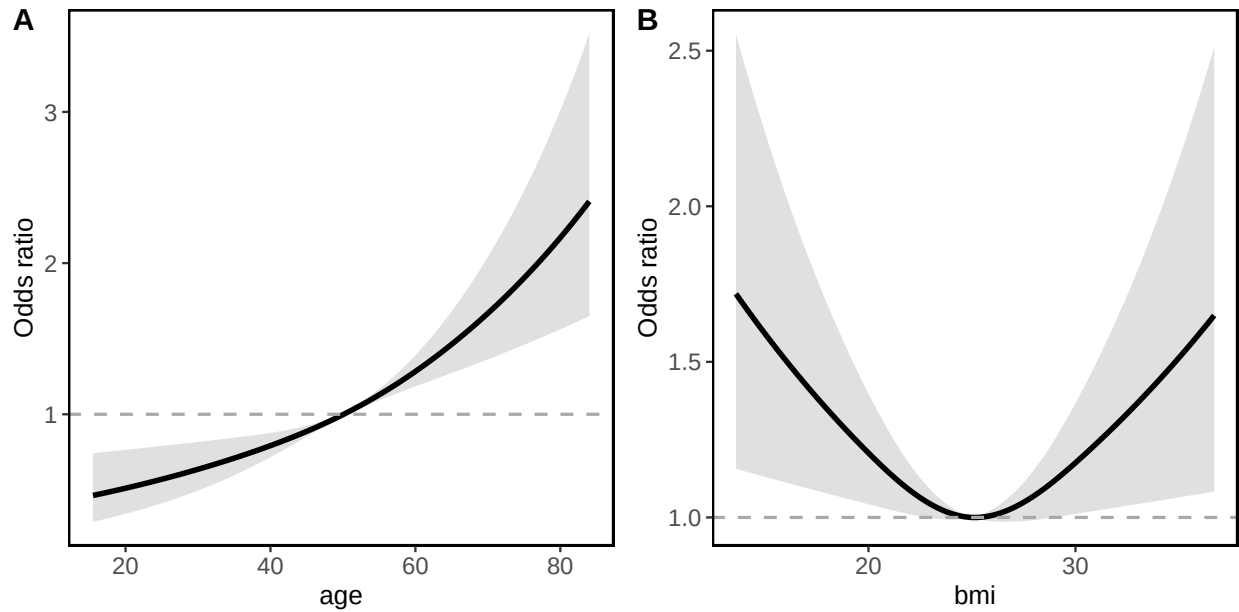
```
## Logistic Regression Model
##
## lrm(formula = majorcomplication ~ rcs(age, 3) + rcs(bmi, 3) +
## sex + smoking, data = data, x = TRUE, y = TRUE)
##
##               Model Likelihood      Discrimination      Rank Discrim.
##               Ratio Test              Indexes          Indexes
## Obs           5000    LR chi2      155.74      R2        0.052    C        0.636
## 0             4169    d.f.          7        R2(7,5000)0.029    Dxy      0.271
## 1             831    Pr(> chi2) <0.0001    R2(7,2078.7)0.069    gamma   0.271
## max |deriv| 3e-07              Brier      0.134    tau-a    0.075
##
##           Coef      S.E.   Wald Z Pr(>|Z|)
## Intercept  -1.8860 0.5600  -3.37  0.0008
## age         0.0219 0.0078   2.80  0.0052
## age'        0.0028 0.0084   0.34  0.7354
## bmi        -0.0552 0.0200  -2.76  0.0059
## bmi'        0.0711 0.0243   2.92  0.0035
## sex=Male    0.0755 0.0774   0.98  0.3293
## smoking=Former -0.0142 0.1031 -0.14  0.8902
## smoking=Current 0.7317 0.0933   7.84 <0.0001
```

```
# Use modelsummary_rms function from rmsMD package to get an output for medical journals
# for spline terms this gives an overall p-value for the association of that variable with outcome
modelsummary_rms(fit)
```

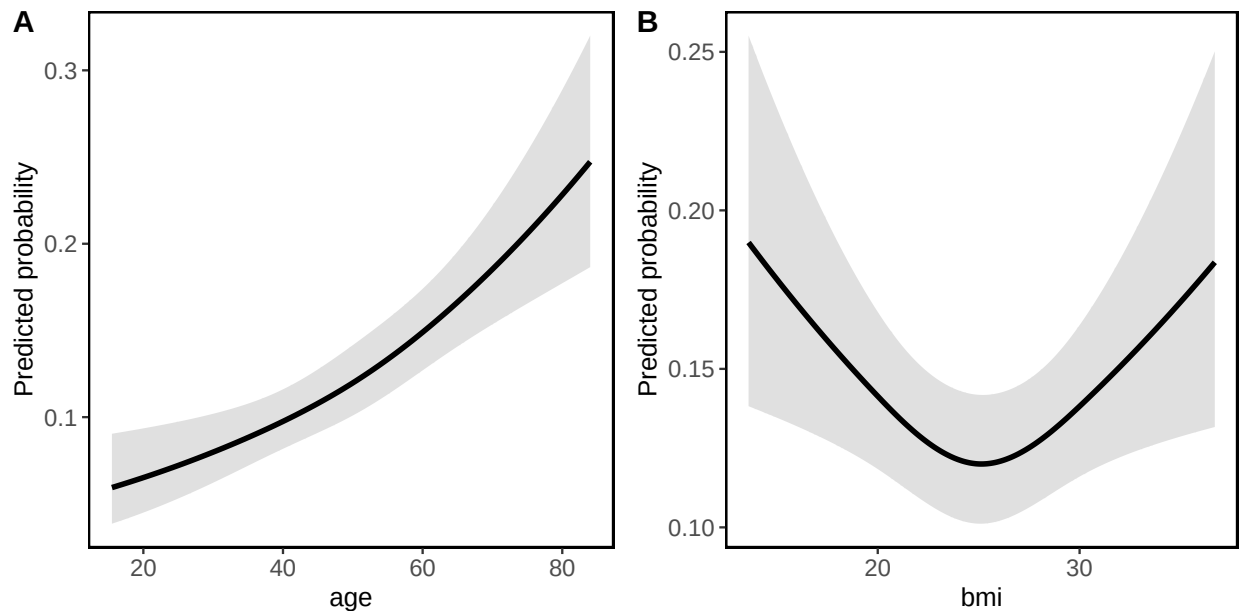
```
##           variable              OR_95CI Pvalue
## 1      sex=Female                Ref      -
## 2      sex=Male 1.078 (0.927 to 1.255) 0.329
## 3      smoking=Never                Ref      -
## 4      smoking=Former 0.986 (0.806 to 1.207) 0.890
## 5      smoking=Current 2.079 (1.731 to 2.496) <0.001
## 6 RCSoverallP: age                LR test <0.001
## 7 RCSoverallP: bmi                LR test 0.015
```

```
# Use ggrmsMD from rmsMD package to plot the splines
# This determines which variables were analysed as RCS, and plots them appropriately
```

```
# As this is a logistic regression it plots OR and 95% confidence interval
# combined = TRUE means that a single combined plot with all spline terms in the model is outputted
ggrmsMD(fit, data, combined = TRUE)
```



```
# for logistic regression models. the lrm_prob argument can be used to plot predicted probabilities
# rather than odds ratios
ggrmsMD(fit, data, combined = TRUE, lrm_prob = TRUE)
```



```
# To assess whether rcs variables are significantly non-linear use anova
# for each RCS term, a p-value for "Nonlinear" is given
# test = "LR" is used to perform a Likelihood Ratio test
anova(fit, test = "LR")
```

```
##                               Likelihood Ratio Statistics      Response: majorcomplication
##
## Factor                Chi-Square d.f. P
## age                   57.40      2   <.0001
##   Nonlinear           0.11       1   0.7359
## bmi                   8.41       2   0.0149
##   Nonlinear           8.29       1   0.0040
## sex                   0.95       1   0.3291
## smoking              88.35       2   <.0001
## TOTAL NONLINEAR      8.38       2   0.0151
## TOTAL                155.74      7   <.0001
```

```
# As expected, BMI, but not age, is significantly non-linear
```

Publication ready outputs using rmsMD

Publication ready tables

The packages `flextable` and `officer` can be used to output results from `modelsummary_rms()` to word documents. These are used in Table 2 of the accompanying manuscript.

```
# modelsummary_rms will output a dataframe. first make this into a flextable
results <- modelsummary_rms(fit)
results_flextable <- flextable(results)

# output to a word document
doc <- read_docx()
doc <- body_add_flextable(doc, results_flextable)
print(doc, target = "Results_of_main_model.docx")

# creating a model which does not use RCS terms to use for comparison
# note this model assumes linear relationships, and incorrectly finds no association
# between bmi and outcome
fit_linear <- lrm(majorcomplication ~ age + bmi + sex + smoking,
  data = data,
  x = TRUE, y = TRUE)
fit_linear # check diagnostics etc
```

```
## Logistic Regression Model
##
## lrm(formula = majorcomplication ~ age + bmi + sex + smoking,
##      data = data, x = TRUE, y = TRUE)
##
##                               Model Likelihood      Discrimination      Rank Discrim.
##                               Ratio Test           Indexes           Indexes
## Obs                5000    LR chi2      147.35      R2          0.049      C          0.633
## 0                  4169    d.f.          5          R2(5,5000)0.028      Dxy         0.267
## 1                  831    Pr(> chi2) <0.0001      R2(5,2078.7)0.066      gamma        0.267
## max |deriv| 1e-08          Brier        0.134      tau-a         0.074
##
##                               Coef      S.E.    Wald Z Pr(>|Z|)
```

```
## Intercept      -3.0831 0.3114 -9.90 <0.0001
## age            0.0244 0.0032  7.52 <0.0001
## bmi           -0.0035 0.0097 -0.36 0.7173
## sex=Male       0.0770 0.0773  1.00 0.3194
## smoking=Former -0.0229 0.1029 -0.22 0.8241
## smoking=Current 0.7242 0.0932  7.77 <0.0001
```

```
results_linear <- modelsummary_rms(fit_linear) # get results dataframe
results_linear_flextable <- flextable(results_linear)
doc <- read_docx() # set up word doc
doc <- body_add_flextable(doc, results_linear_flextable) # add results
print(doc, target = "Results_of_model_with_linear_assumption.docx")
```

Publication ready plots

Using `ggrmsMD()` from `rmsMD` to make Figure 2 in the accompanying manuscript.

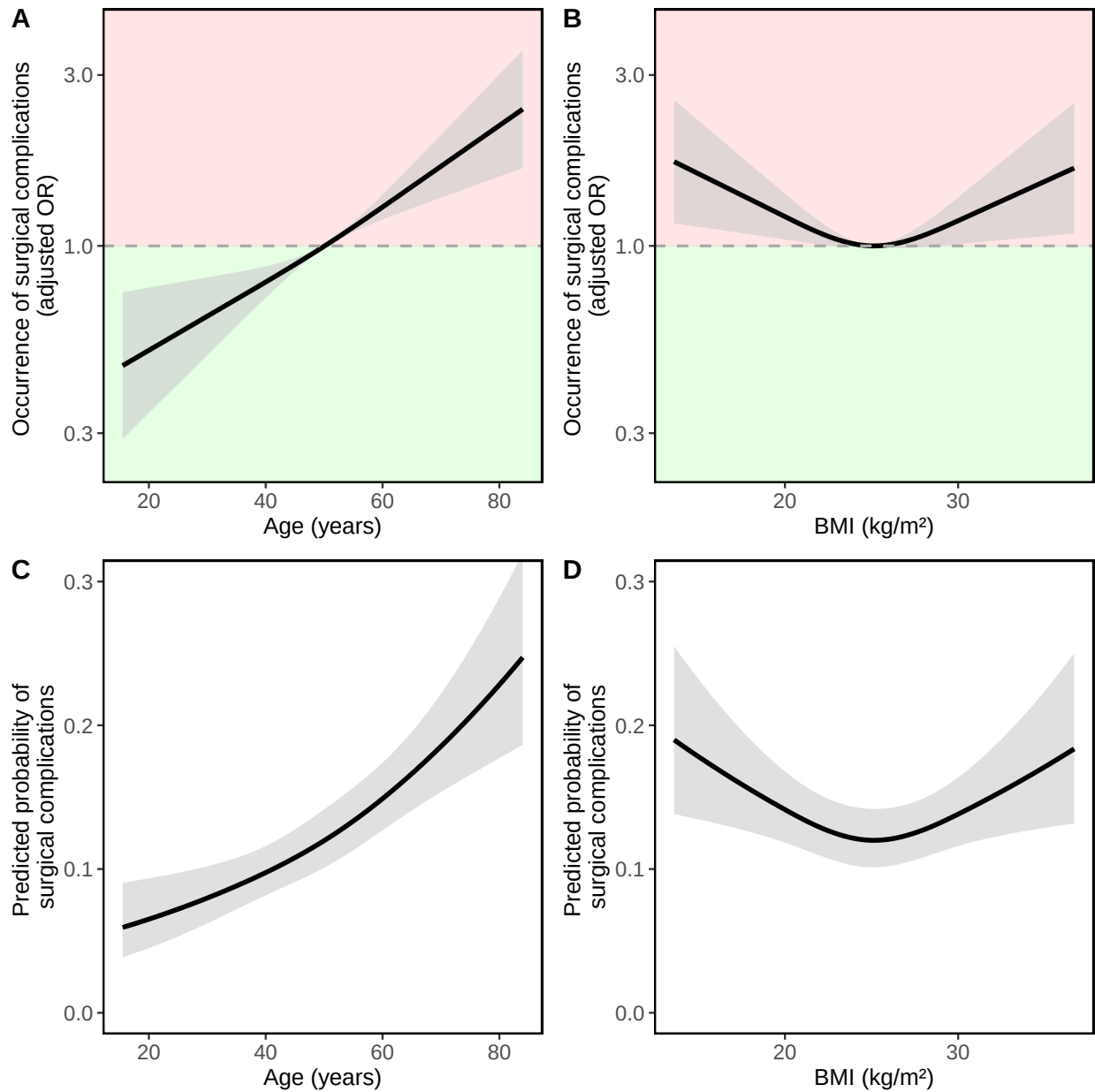
```
# plot adjusted odds ratio. note the y axis is plotted here on a log-scale rather than
# a linear scale. The shade_inferior argument is used to shade either side of the no
# effect line (OR = 1) to give a visual cue for which side represents inferior outcome
# note that by setting combined = FALSE, the output is a list of plots, one for each RCS term
plots_OR <- ggrmsMD(fit, data, combined = FALSE,
  shade_inferior = "higher",
  ylab = "Occurrence of surgical complications \n(adjusted OR)",
  xlab = list(age = "Age (years)", bmi = "BMI (kg/m²)"),
  log_y = TRUE,
  ylim = c(0.25, 4)
)

# plot the predicted probability
plots_prob <- ggrmsMD(fit, data, combined = FALSE,
  ylab = "Predicted probability of \nsurgical complications",
  xlab = list(age = "Age (years)", bmi = "BMI (kg/m²)"),
  lrm_prob = TRUE,
  ylim = c(0,0.3)
)

# combine the plot lists and making a single multipanel figure
plotlist <- c(plots_OR, plots_prob)

# using plot_grid from `cowplot` package
plots <- plot_grid(plotlist = plotlist, labels = "AUTO", align = "vh")

# to view the plot
plots
```



```
# to save as pdf
ggsave(file = "Case_study_RCS_plot.pdf", plot = plots, width = 8, height = 8)

# to save as jpg
ggsave(file = "Case_study_RCS_plot.jpg", plot = plots, width = 8, height = 8)
```